

Class 30006 – Financial Markets and Institutions  
Università Commerciale Luigi Bocconi  
Fall 2025

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# The Money markets (Chapter 11)



# Outline of Lecture

## 1. Money markets

- Definition
- Purpose
- Players

## 2. Money markets instruments

- Treasury Bills
- Federal Funds
- Repurchase Agreements
- Negotiable Certificates of Deposit
- Commercial Paper
- Eurodollars

# Definition

- ✓ We call it “**money market**” (MM), because the securities in these market are (see front slide)
  - **short term**
  - with **high liquidity**
- ✓ so, they are close to being money ...
- ✓ ... but money is NOT actually traded
  - (≠ foreign exchange market!)
- ✓ Money market instruments are generally used:
  - to fill short-term financial needs (supply side)
  - to temporally “warehouse” surplus funds (demand side)
- Optional: during the Financial Crisis of 2008 many MM instruments lost their liquidity, this situation created panic (we will see it in specific lecture)

# Definition & chars

Three key characteristics:

**1. Usually sold in large denominations**

**2. Low default risk**

**3. Original (=at issue) maturity of one year or less**

- although vast majority matures in less than 120 days (many are even *overnight* “o/n”)
- note: original maturity  $\neq$  residual maturity! Ex.: a 3-year bond with 2 months left to maturity (residual maturity) is NOT a money market instrument

✓ Most money market transactions are made through over-the-counter markets (OTC)

- OTC Vs Organized Exchange (OE) ...
- ... but difference recently blurred

# Purpose (lenders)

## Demand

- ✓ For *buyers* of MM instruments (lenders):
  - warehouse surplus funds for short periods of time
  - they accept low returns
  - more interested to **liquidity**: act quickly if other investments become available
  - Example: over the last years, Apple has held around \$50-100bn in “cash, cash equivalents and marketable securities”
  - Not just Apple: top 30 companies have >\$1tn in cash
- ✓ Clearly, they do not just put all this \$ in a bank account ...  
... they use **money markets**

# Purpose (borrowers)

## Supply

For *sellers* of MM instruments (borrowers):

✓ **low-cost** source of temporary funds

- again, not for large capital projects with high return

✓ cash management

- the **timing** of cash inflows and outflows are not well synchronized
- For ex.:
  - Candy company pays salaries every month, but sells most of its product on Halloween
  - Ice cream company sells most during summer, but pays salaries every month
  - Gov. collects taxes at fixed dates, but pays gov. expenditure every month
  - Banks need to adjust reserves overnight at Central Bank
  - etc...

# MM Players

**Sellers**  
(Borrowers)

**Buyers**  
(Lenders)

**Government  
(Treasury)**

**Banks**

**Money Market  
Mutual Funds**

**Corporations**

Insurance Companies

**Central  
Bank**

Pension Funds

Ex.: Gov. issues Treasury bills (T-bills) to finance short term expenditure  
For details of use by each subject: see book and next slide

# Players

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| Player                           | Name in the US           | Activity                                                                                                |
|----------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Government</b>                | U.S. Treasury Department | issues T-bills, always a demander, never a supplier of funds                                            |
| <b>Central Bank</b>              | Federal Reserve System   | buys/sells treasury securities to control the money supply in the economy                               |
| <b>Commercial Banks</b>          |                          | hold treasury securities, issue CDs, Fed funds, bank's acceptances, repos. Trade on behalf of customers |
| <b>Businesses</b>                |                          | buy and sell various short-term securities as a regular part of their cash management                   |
| <b>Insurance Companies</b>       |                          | maintain liquidity need for unexpected demands (e.g., tornadoes in US in 2015)                          |
| <b>Pension funds</b>             |                          | temporary storage to have money readily available for investment opportunities                          |
| <b>Money market mutual funds</b> |                          | invest in large-denomination money market securities on behalf of many individual small investors       |

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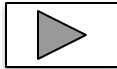
# Money Market Instruments

Money Markets instruments at glance:

1. Treasury Bills
2. Federal Funds
3. Repurchase Agreements
4. Negotiable Certificates of Deposit
5. Commercial Paper
6. Eurodollars

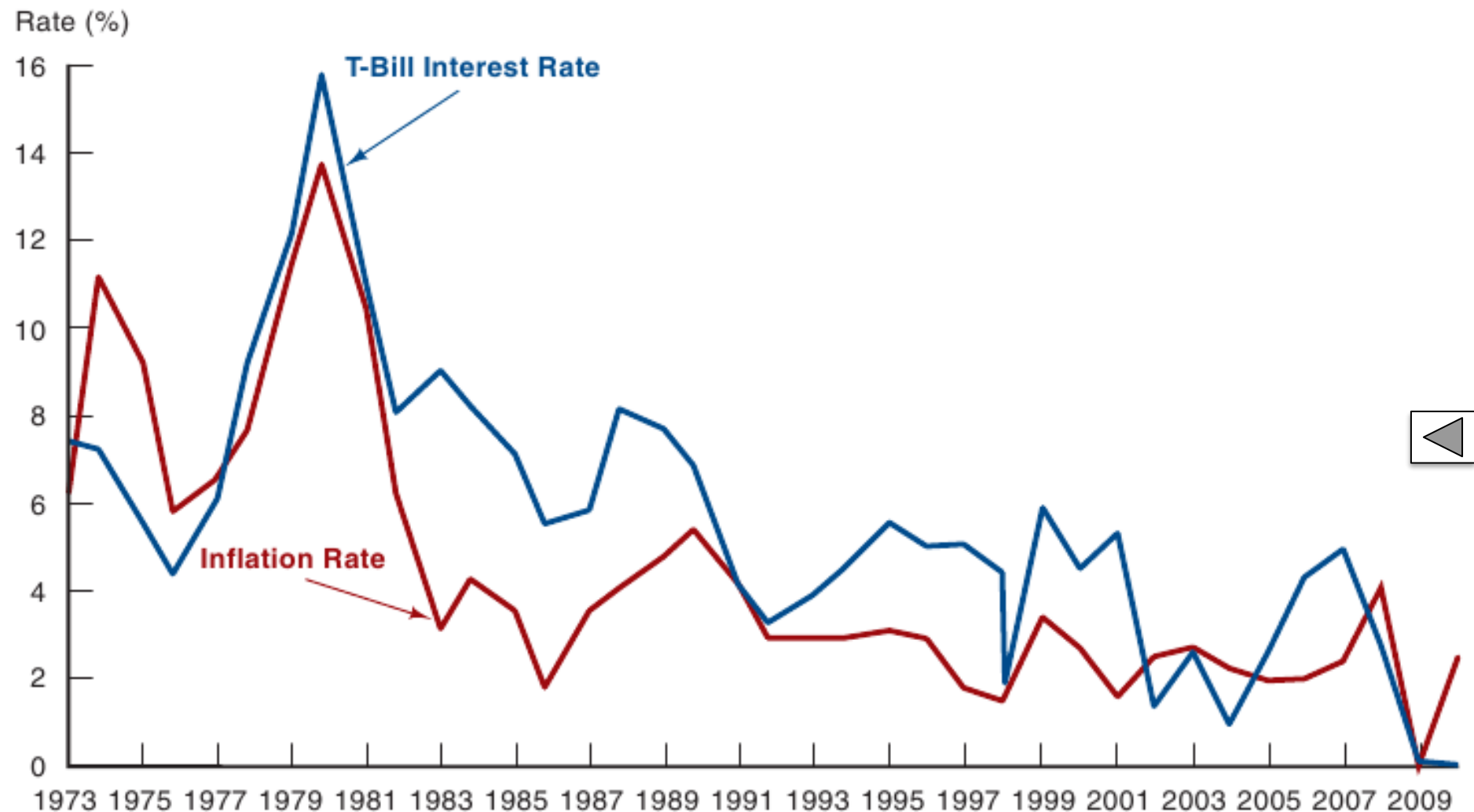
# Money Market Instruments:

## Treasury Bills

- ✓ T-bills are **short-term** obligations issued by the U.S. Government
  - Maturity: 4-, 13-, 26-, and 52-weeks ( $\leq 1$  year)
- ✓ T-bills are *discount bonds*
  - discounting is common to S.T. securities because historically there was little time to collect coupons
- ✓ One of most liquid financial markets
  - practically **riskless**, with liquid, deep market, but with low return: earnings often barely above (and sometimes below) inflation 
- ✓ Sold in auction. Two types of bidders:
  - noncompetitive bidding: submit amount only, buy *at any* price
  - competitive bidding: submit amount *and* price
- ✓ Book entry security
  - financial asset whose ownership is recorded and tracked electronically rather than through physical certificates
- ✓ Minimum amount: \$100

# Money Market Instruments: Treasury Bills

Earnings often barely above (and sometimes below) inflation



# Treasury Bill Auctions

✓ T-bills are sold in **auctions**:

- they are auctioned to the *primary dealers* (about [23 banks](#)): example of primary market
- Treasury announces how many and what kind of T-bills

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- bidders only provide **only** amount, **not** price
- Treasury automatically accepts all these bids
- noncompetitive bidders will pay the price determined by the competitive bidders, but are **guaranteed** the amount

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## ✓ In a **competitive** *bid*:

- dealers put in quantity **and** price
- the bids are accepted in ascending order of yield (descending of price) until offering amount is reached
- Treasury will then sell T-bills at the highest yield (**lowest** price) among the **accepted** bids

# Treasury Bill Auctions: example

- ✓ The Treasury auctioned \$2.5 billion par value  
Maturity: 91 days (=13 weeks)
- ✓ The following bids were received:
  - a. \$750 million in **noncompetitive** bids
  - b. the following **competitive** bids:

| Bidder | Bid Amount    | Bid Price | Int. rate <sup>(1)</sup> |
|--------|---------------|-----------|--------------------------|
| 1      | \$500 million | \$0.9940  | 2.37%                    |
| 2      | \$750 million | \$0.9901  | 3.92%                    |
| 3      | \$1.5 billion | \$0.9925  | 2.97%                    |
| 4      | \$650 million | \$0.9936  | 2.53%                    |
| 5      | \$600 million | \$0.9939  | 2.41%                    |

(1): computed as a discount rate  $= \frac{1 - P}{1} \times \frac{360}{91}$

**Q:** Who will receive T-bills, what quantity, and at what price?

# Treasury Bill Auctions: example

**A:** 1<sup>st</sup>: all non-competitive bidders receive their requested amount: \$750M



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2<sup>nd</sup>: in **competitive bids**

- the **residual** amount is to be allotted:  $\$2,500M - \$750M = \$1,750M$
- the T-bills are allotted to the highest bidders until all T-bills are sold ...
- ... the Treasury accepts, for the amount ( $\$1,750M$ ), the following bids:

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| <u>4</u> | <u><b>\$650 million</b></u> | <u><b>\$0.9936</b></u> | <u><b>2.53%</b></u>      |
| 3        | \$1.5 billion               | \$0.9925               | 2.97%                    |
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only these 3  
bids are  
accepted:  $\Sigma =$   
\$1,750M

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# Treasury Bill Auctions: example

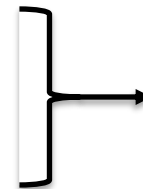
- ✓ In practice:
  - re-order bids in ascending int. rate (or descending price)
  - stop when full amount is allotted
- ✓ Note:
  - both the competitive and noncompetitive bidders pay the *highest yield/lowest bid price*: \$0.9936
  - all T-bills were sold ...
  - ... and all other bidders in **c.b.** receive no allotment
- ✓ In this example, sum of accepted bids **in n.c.b.'s** = residual amount
- ✓ What if sum of accepted bids **in n.c.b.'s** > residual amount ?

# Treasury Bill Auctions: example

**Q: Special case:** sum of accepted bids > residual amount

- In the previous example, assume the following **competitive bids**:

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| 3        | \$1.5 billion        | \$0.9925        | 2.97%                    |
| 2        | \$750 million        | \$0.9901        | 3.92%                    |



only these 3 bids  
are accepted,  
but sum (\$1,950M)  
is **greater** than  
residual (\$1,750M)

(1): computed as a discount rate

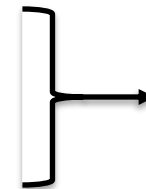
# Treasury Bill Auctions: example

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(1): computed as a discount rate



only these 3 bids are accepted, but sum (\$1,950M) is **greater** than residual (\$1,750M)

**A:** marginal bid is “accepted in less than the full amount you want if your bid is equal to the high discount rate”

- So marginal bidder (no. 4) obtains less than requested: only \$650M
  - see: <https://www.treasurydirect.gov/research-center/history-of-marketable-securities/bills/t-bills-indepth>

# Treasury Bill Auction

## ✓ Summing up:

- **Non-competitive** bids determine **quantities** for each of the two auctions, not price
- **Competitive** bids determine **price** applied to all bidders ( $P^*$  = minimum price among accepted competitive bids)

## ✓ Note that to ensure competitiveness

- in a single auction, a bidder can:
  - buy up to \$10 million in T-bills by non-competitive bidding
  - or
  - buy up to 35% of the initial offering amount by **competitive** bidding (see: <https://www.treasurydirect.gov/research-center/history-of-marketable-securities/bills/t-bills-indepth>)

## ✓ After buy, the primary dealers may then sell the T-bills to the public (*secondary market*)

- they act as the *market maker* for the security

# Treasury Bill Auction Results

T-bills are often quoted with two different rates:

✓ *discount rate* 
$$i_{discount} = \frac{F - P}{F} \times \frac{360}{n}$$

- in the denominator you have F rather than P: it is a bit misleading, but it is easy to calculate even without calculator

✓ *investment rate* 
$$i_{yt} = \frac{F - P}{P} \times \frac{365}{n}$$

where:

$n$ : days to maturity

$F$ : face value

$P$ : purchase price

- ✓ *investment rate more accurate measure of earnings ...*
- *denominator is actual price paid*
  - *365 is actual number of days*

# Treasury Bill Auction Results

| Security Term | Issue Date | Maturity Date | Discount Rate | Investment Rate | Price per \$100 | CUSIP     |
|---------------|------------|---------------|---------------|-----------------|-----------------|-----------|
| 28 days       | 5/16/2013  | 6/13/2013     | 0.01          | 0.01            | 99.999222       | 912796AL3 |
| 91 days       | 5/16/2013  | 8/15/2013     | 0.045         | 0.046           | 99.988625       | 912796AX7 |
| 182 days      | 5/16/2013  | 11/14/2013    | 0.08          | 0.081           | 99.959556       | 912796AE9 |
| 28 days       | 5/9/2013   | 6/6/2013      | 0             | 0               | 100             | 912796AJ8 |
| 91 days       | 5/9/2013   | 8/8/2013      | 0.04          | 0.041           | 99.989889       | 912796AV1 |

Source: <http://www.treasurydirect.gov/RI/OFBills>.



# Fed Funds

- ✓ **Fed Funds:** short-term funds
  - exchanged (loaned/borrowed) between *financial institutions* (e.g.: banks), not Fed Gov.
  - only some fin. Intermediaries (“banks”) can participate
  - name (“Fed”) because funds held at Federal Reserve
- ✓ Banks lend/borrow federal funds usually ***overnight*** (=less than 1 day!)
  - **unsecured**
  - direct communication between fin. institutions
- ✓ The “banks” use of federal funds for:
  - satisfying reserve requirements set by Fed
  - clearing of transactions (payments)

# Fed Funds

- ✓ The interest rate is called *target/intended Fed Funds rate* because that's what the Fed wants to see
  - Monetary policy implemented on this market: when you read “*The Fed has increased the interest rate to 1.25%*” they mean the **Fed Funds rate**
  - Fed controls *indirectly* Fed funds rate by purchasing or selling securities (using federal funds) :
    - if Fed **buys** securities, banks receive funds that deposit as reserves at Fed: supply of reserves increases and thus Fed Funds **rate** ↓
    - the Fed **sells** securities if want Fed Funds **rate** ↑
- ✓ The *effective* rate
  - determined by demand and supply between banks (weighted avg int. rate) ...
  - ... but it moves around target (Fed target rate: see later lecture)

# Target Fed Funds Rate



# Target and Effective Fed Funds Rate



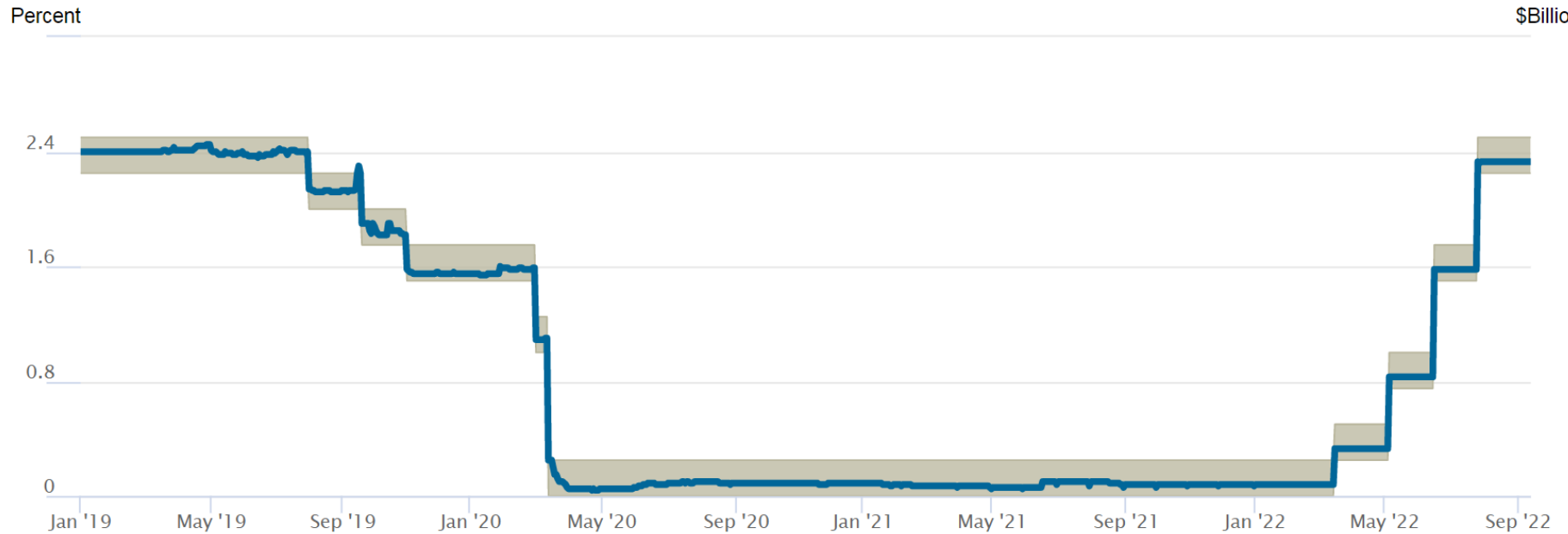
# More recently...

## FEDERAL FUNDS CHART

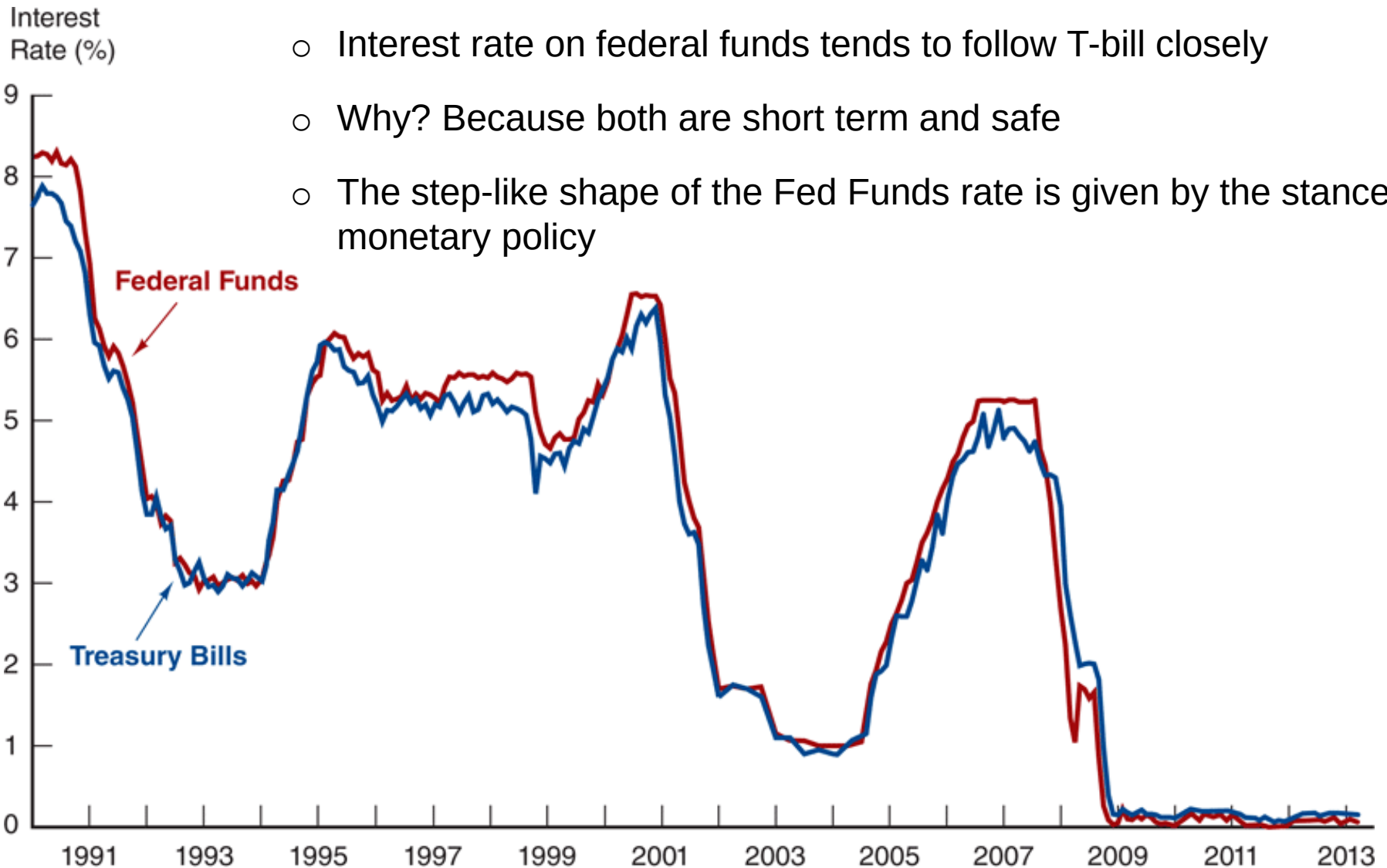
1m 3m 1y All

From Jan 1, 2019 To Sep 14, 2022

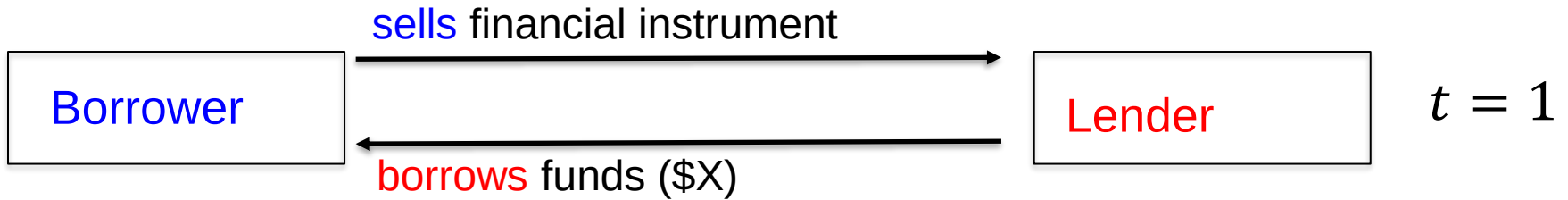
● Target Rate/Range — EFR — Volume  
Percent \$Billions



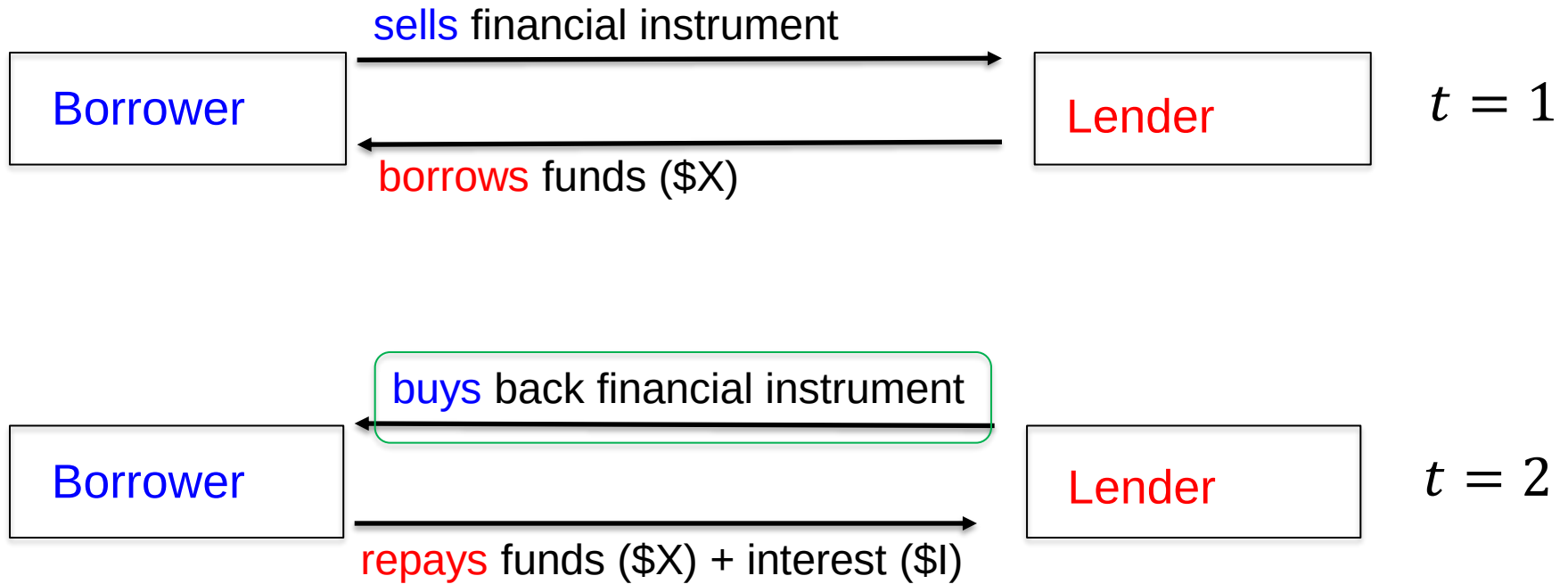
# Effective Fed Funds and T-Bill Rates



# Repurchase Agreements (Repo)



# Repurchase Agreements (Repo)



- ✓ In every bond, issuer (=borrower) sells bond to borrow
- ✓ in addition to selling the financial instrument, here issuer
  - commits to repurchase the same financial instrument ... at higher price



# Repurchase Agreements (Repo)

Called “Repo”

- ✓ It is an agreement where a financial instrument is sold and bought back (repurchased) at a specified future date and at a determined price
  - mostly short term (usually 3 to 14 days)
  - *example*: the borrower (say, a bank) sells **financial instrument (ex.: T-bills)** for \$X to the lender and agrees to buy them back at a future date for  **$\$Y(=\$X+\$I)>\$X$**
- ✓ Quite similar to a collateralized short-term loan
  - if the borrower does **not** buy back, the lender can sell the *collateral* (typically *T-bills*)
  - note that Repo market collapsed for some time in 2007-08 because collaterals lost value and investors lost trust (optional: "the run on repos")

# Repurchase Agreements (Repo)

Repo-rate:

$$\text{Repo Rate} = \frac{\$Y - \$X}{\$X}$$

- ✓ The difference between  $\$X$  and  $\mathbf{P}$ , the market price of **collateral**, is called **haircut**:

$$\text{haircut} = \$X - P$$

- ✓ often the collateral consists of government bonds (e.g.: T-bills): highly liquid, stable securities (say 2% haircut)

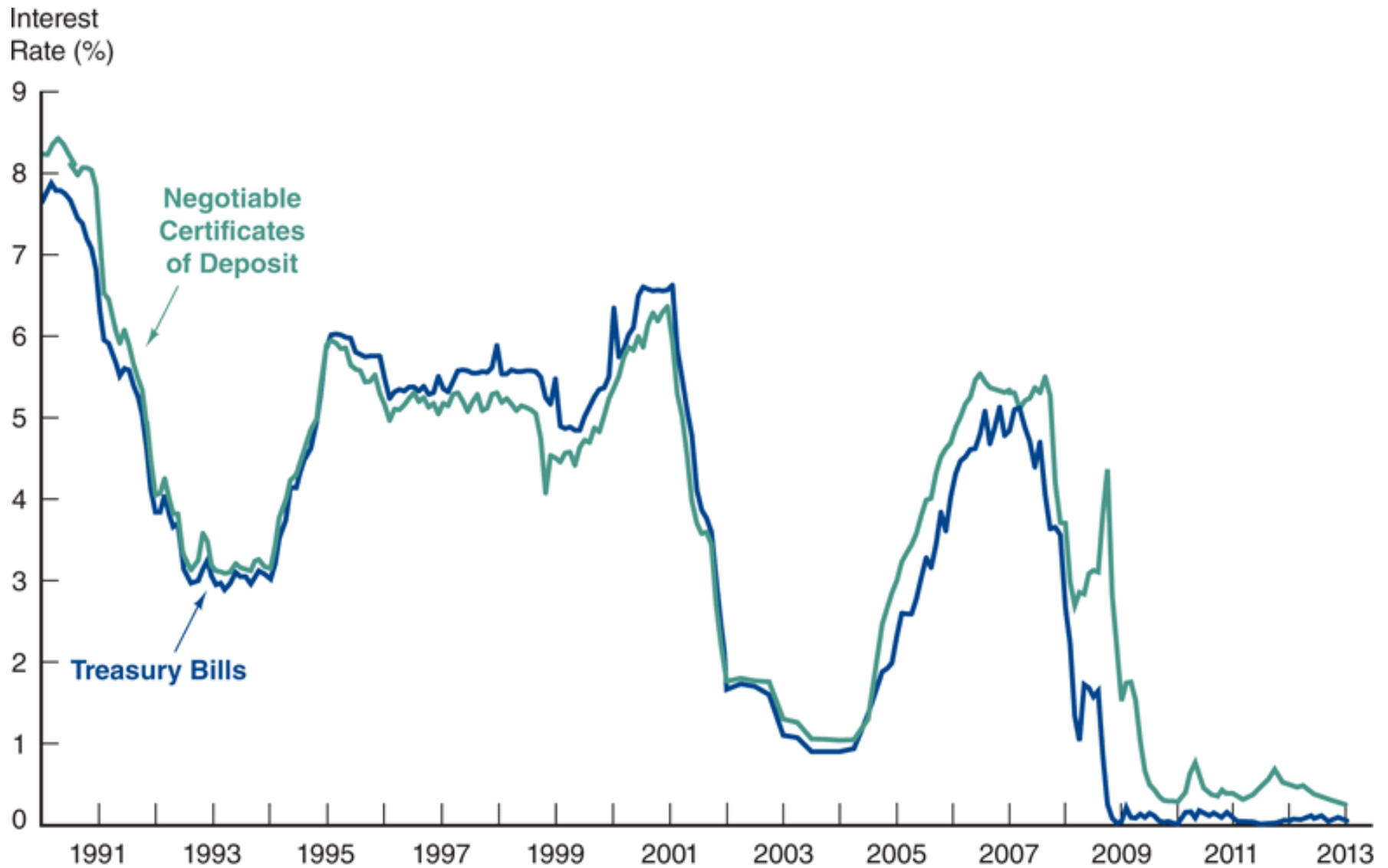
# Repurchase Agreements

- ✓ The set-up makes a repo agreements essentially a **short-term collateralized loan**
- ✓ This means that the lender is *secured*
- ✓ The repos are normally low risk because S-T and are backed by a liquid collateral. But losses can happen!
- ✓ *Differences with Fed Funds:*
  - main: repos are secured loans, Fed Funds are unsecured loans
  - another: *only some fin. intermediaries* allowed to trade Fed Funds, with repos, *also non-banks* can participate

# Negotiable Certificates of Deposit (CD's)

- ✓ A **bank**-issued deposit that specifies
  - the **interest** rate (fixed)
  - the **maturity** date (fixed)
- ✓ It is a **term deposit**, *i.e.* you cannot withdraw before maturity (as opposed to **demand deposit**)
- ✓ At maturity bearer of the certificate is entitled to the principal and interest
- ✓ Characteristics
  - large denominations, ranging from \$100K to \$10M
  - buyers/lenders: wealthy individuals and institutional investors (4 ex. pension funds)
  - maturity between 14 days and 4 months
- ✓ One of the most important money market instruments
  - traded in large volumes, second to T-bill only

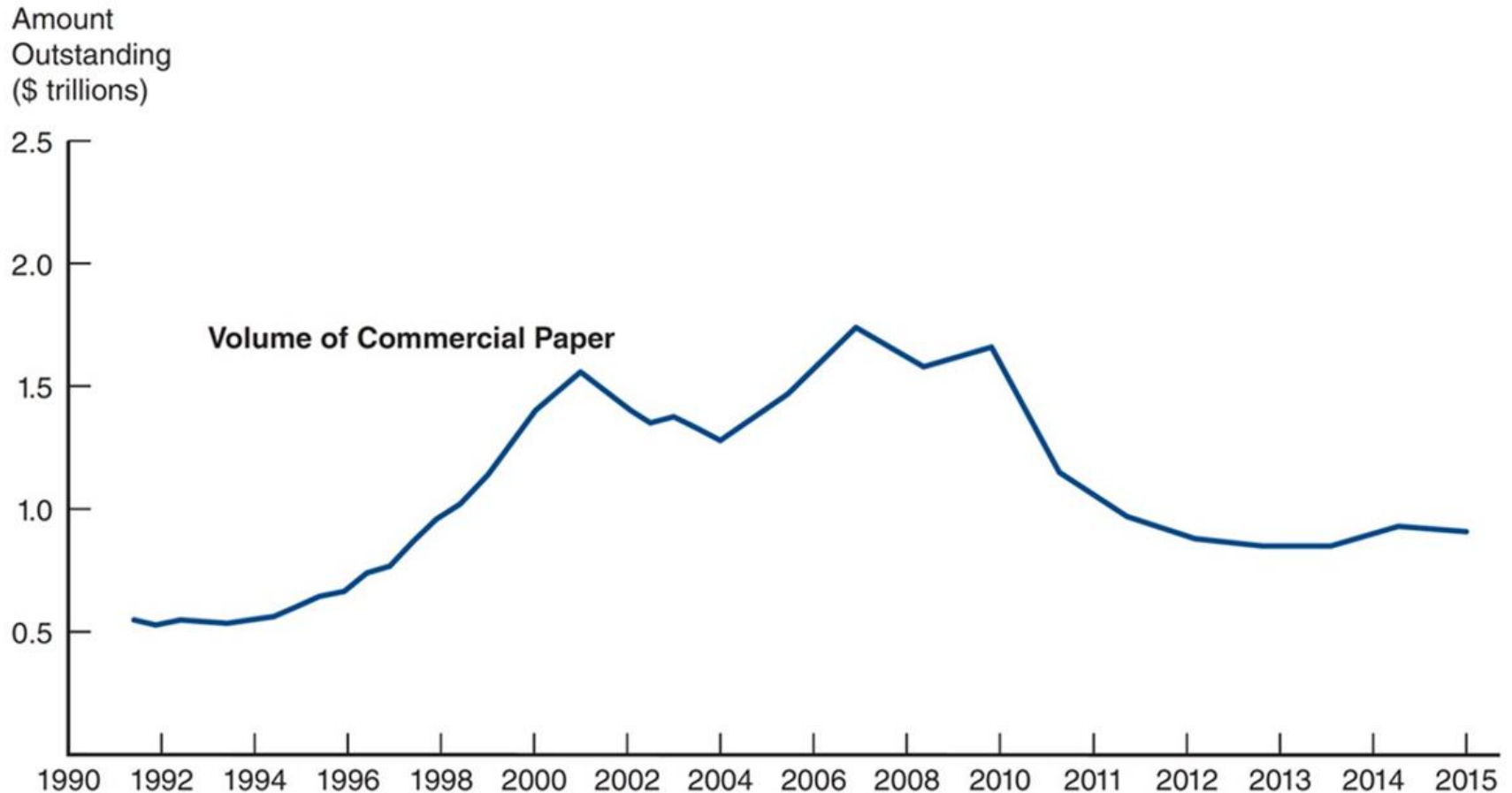
# CD Rates and T-bills



# Commercial Paper (CP)

- ✓ **Unsecured promissory notes** (a promise to pay), issued by **corporations** with maturity  $\leq 270$  days
- ✓ Sold on a discount basis, like T-bills
- ✓ Since CP is normally **unsecured**
  - unlike bank deposits, CPs are *not insured (only by collateral if exists)*
  - interest rate reflects *firm's* level of *risk*
  - issued only by largest and trustworthy corporations (also banks)
- ✓ Often sold directly from issuer to buyers
  - *essentially no secondary market*
  - purchasers at first were only banks. Nowadays also others (large insurance co.'s, non-financial businesses, etc...)
- ✓ The use of CP increased significantly in the early 1980s because of the rising cost of bank loans

# Commercial Paper



Source: <http://www.federalreserve.gov/releases/cp/yrend.htm>

But see latest data [here](#)

# Eurodollars

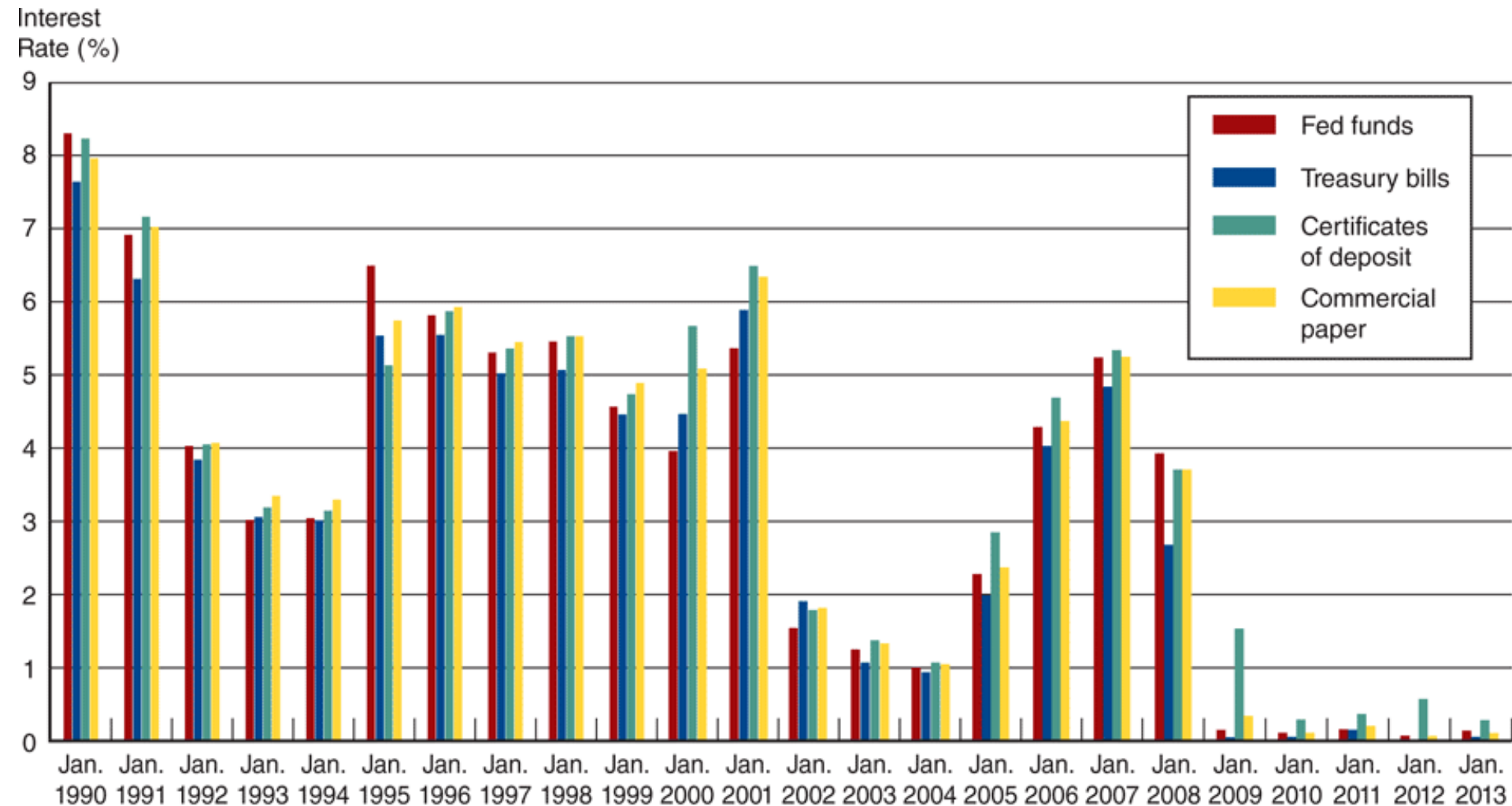
- ✓ Eurodollars: \$ denominated deposits held in foreign banks
  - name: originally only in European banks (London)
  - history: USSR had lot of USD in US banks: risk of seize!!
  - nowadays: many contracts out of US call for payment in USD
  - short term **interbank deposits**, often *overnight* (o/n): alternative source of short-term funding for banks
  
- ✓ Eurobonds: following the Eurodollar logic), they are bonds issued outside of the U.S. but typically denominated in **USD** [not in **Euros**!]
  - usually long-term debt instrument



# Eurodollars: Libor

- ✓ The most important interbank market of Eurodollars **was** **LIBOR (London interbank offer rate)**:
  - *“At what rate could you borrow funds, were you to do so by asking for and then accepting inter-bank offers [...] just prior to 11 am?”*
  - nowadays Eurodollar market is no longer just London, but worldwide
  - in many currencies: \$, €, ¥, £
  - various maturities: o/n, 1M, 3M, 6M, 12M
- ✓ After manipulation scandals (see optional slide)
  - ✓ LIBOR progressively phased out and replaced
  - ✓ last USD LIBOR rates were published at the end of Sep 2024

# A Comparison of Rates: all have VERY similar levels!



# Money Market Securities and Their Depth

| Money Market Security              | Issuer                           | Buyer                              | Usual Maturity      | Secondary Market |
|------------------------------------|----------------------------------|------------------------------------|---------------------|------------------|
| Treasury bills                     | U.S. government                  | Consumers and companies            | 4, 13, and 26 weeks | Excellent        |
| Federal funds                      | Banks                            | Banks                              | 1 to 7 days         | None             |
| Repurchase agreements              | Businesses and banks             | Businesses and banks               | 1 to 15 days        | Good             |
| Negotiable certificates of deposit | Large money center banks         | Businesses                         | 14 to 120 days      | Good             |
| Commercial paper                   | Finance companies and businesses | Businesses                         | 1 to 270 days       | Poor             |
| Banker's acceptance                | Banks                            | Businesses                         | 30 to 180 days      | Good             |
| Eurodollar deposits                | Non-U.S. banks                   | Businesses, governments, and banks | 1 day to 1 year     | Poor             |

# LIBOR FIXING SCANDAL (optional)

- ✓ Since their inception, Libor rates played a fundamental role in the financial world: they were used as reference borrowing costs in financial contracts worth trillions of dollars in total
- ✓ However, in 2012, the Libor draw much attention for being at the center of a **scandal**:
  - it emerged that banks and traders had been manipulating it for years: they were colluding to “fix” the LIBOR
  - Why? To earn money on interest rate swaps. These are basically bets that (variable) interest rates like LIBOR will go down versus some fixed rate. If you can control the variable rate, it’s an easy bet!
- ✓ How? Only 18 banks set the \$-LIBOR. If a few collude, they can manipulate the average
- ✓ In 2017, decided to discontinue it, with the official deadline to replace it being 2021. Since 2022 LIBOR no longer used as benchmark:
  - replaced with a host of different rates that were more suited to represent borrowing costs in different jurisdictions